

SIGN LANGUAGE TRANSLATOR

Sign Language Interpreter and Learning Management System

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Abstract

The Sign Language Interpreter and Learning Management System is a Python and MySQL-based application designed to reduce communication barriers by organizing sign-language gestures and their meanings. Users can search for signs, view translations, maintain translation history, access learning modules, and track learning progress. Administrators can manage gesture, translation, and learning-module records.

APPLICATION

Python

DATABASE

MySQL

USER FEATURES

Search • Learn • Track

DBMS

Keys • Joins • Triggers

Problem Statement

Communication becomes difficult when people do not understand sign language.

Existing learning methods are distributed across books, dictionaries, videos, websites, and general translation applications.

Users may lack a centralized way to search signs, retain translation history, and track learning progress.

Administrators may not have an efficient way to manage gesture and translation information.

Problem definition: develop a centralized database-driven platform that combines sign information with user and learning management.

Motivation

Accessibility

Make sign-language information more organized and accessible.

Centralization

Bring gesture meanings, translations, history, learning modules, and progress into one platform.

Learning Support

Help users access structured learning content and monitor their progress.

Practical DBMS

Demonstrate relationships, normalization, SQL queries, joins, triggers, stored procedures, and reporting.

Objectives

- 1 Provide user registration and secure login.
- 2 Enable gesture search and display corresponding meanings/translations.
- 3 Maintain translation history and user learning progress.
- 4 Provide learning modules and administrator management for gesture and learning content.
- 5 Use a relational MySQL database to organize and retrieve system information efficiently.

Literature Survey

The supplied project material describes the existing landscape but does not provide individual research-paper citations, named datasets, or benchmark values. The table below therefore records the literature themes explicitly supported by the source rather than inventing authors or results.

Literature / Approach	Method / Resource	Finding described in source	Limitation
Traditional learning	Books & dictionaries	Provide sign-language information	No centralized progress/history management
Multimedia learning	Videos & websites	Offer visual and online learning content	Information is distributed
General translation apps	Translation applications	Support translation-related use cases	Not described as a full learning-management system
Proposed direction	Python + MySQL platform	Centralized sign, user & learning data	Computer-vision recognition is future work

Research Gap

Existing methods described in the project material are not presented as a single centralized platform.

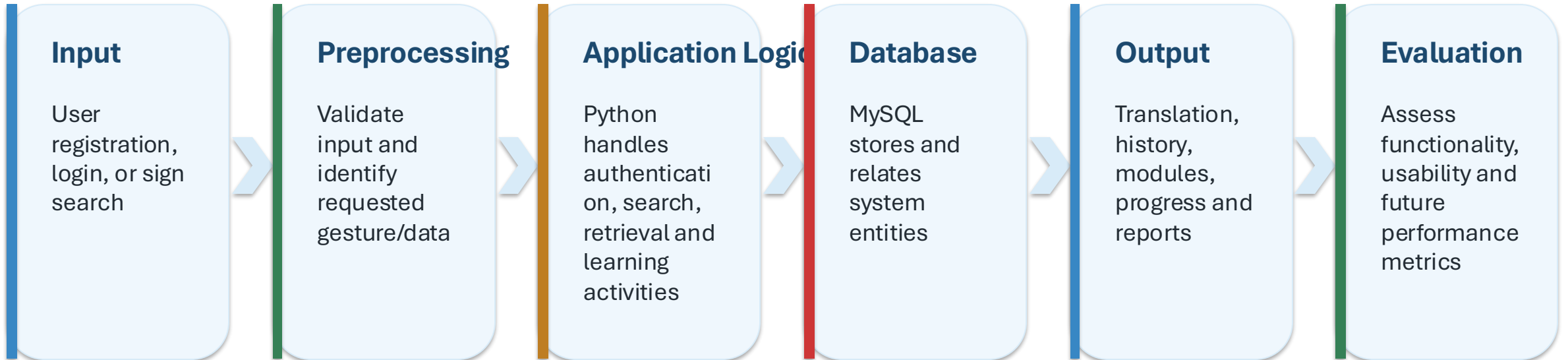
Personalized learning progress and translation history are identified as missing capabilities.

Structured management of gesture, translation, and learning-module data is needed.

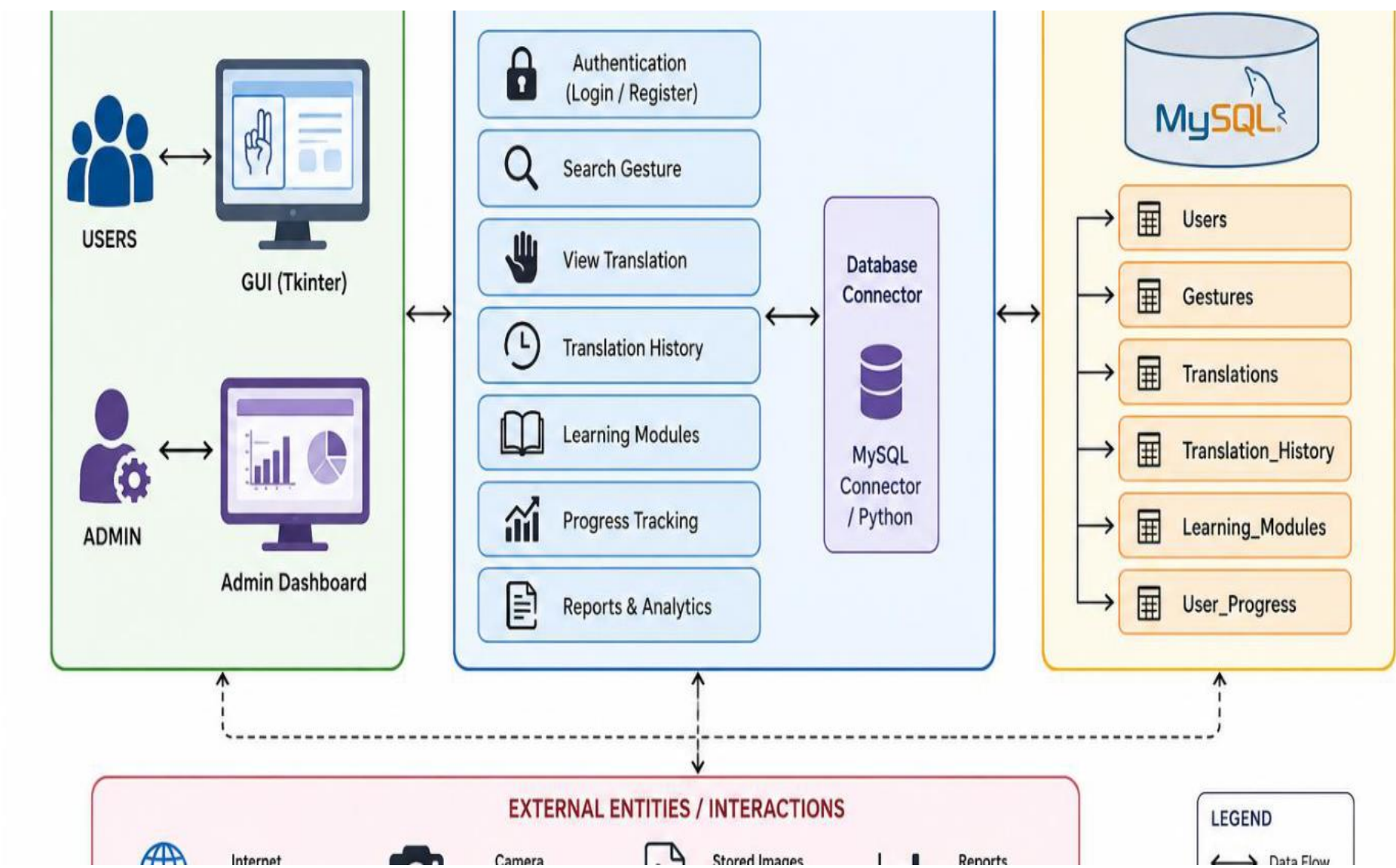
The supplied project does not yet document automatic camera-based sign recognition.

Gap addressed: integrate sign information with user, translation-history, learning-module, and progress management in one database-driven application.

Proposed Methodology



System Architecture



Technologies Used

Programming Language

Python 3.x

User Interface

Python Tkinter

Database

MySQL

Database Connectivity

MySQL Connector/Python

Database Tool

MySQL Workbench / XAMPP

Development Tool

Visual Studio Code

Operating System

Windows/Linux compatible environment

Dataset / Input Details

Dataset name: Not specified in the supplied project document.

Source: Sign-language reference image/content is shown in the project material.

Stored entities: Users, Gestures, Translations, Translation History, Learning Modules, User Progress.

Input parameters: registration/login details and gesture search/input.

Output/target: gesture meaning/translation, learning content, history, progress and reports.

Train/test/validation split: Not applicable to the current database-driven implementation; automatic ML recognition is future work.



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Implementation – Module 1

USER & AUTHENTICATION

Purpose

- Register and log in users

Input

- User credentials

Processing

- Authentication and validation

Output

- User access to the platform

GESTURE SEARCH

Purpose

- Find sign-language gestures

Input

- Search/gesture query

Processing

- Retrieve matching data from MySQL

Output

- Gesture meaning / translation

DBMS FOUNDATION

Purpose

- Organize application data

Technique

- Primary/foreign keys, constraints, normalized relational tables

Output

- Consistent stored data

Implementation – Module 2

TRANSLATION & HISTORY

- Display translations
- Maintain translation history
- Store user-related translation records
- Retrieve historical activity from MySQL

LEARNING MODULES

- Provide structured learning content
- Allow users to access learning modules
- Store module information in MySQL
- Support future addition of learning content

PROGRESS TRACKING

- Record user progress
- Retrieve progress information
- Support learning management
- Enable report generation

Implementation – Module 3

ADMIN MANAGEMENT

- Admin login
- Add gesture records
- Update gesture/translation data
- Delete outdated records
- Manage learning modules

SQL & DATABASE OPERATIONS

- Relationships
- Normalization
- SQL queries and joins
- Triggers
- Stored procedures

REPORTING

- Generate reports from stored data
- Combine user, history and progress information
- Present organized information for management

Experimental Setup

Component	Current setup / status
Hardware	Computer with at least 4 GB RAM (requirement stated in source)
Software	Python 3.x, MySQL Server, MySQL Workbench/XAMPP, VS Code
Database	MySQL relational database
Connectivity	MySQL Connector/Python
Experiment type	Functional database/application testing
Evaluation	Functionality, usability and performance; numerical benchmark values not provided

Results

Based on the supplied project document, the current result is a designed and specified database-driven platform supporting the following functions:

Authentication

Registration and login

Translation

Gesture search and translation display

History

Translation-history management

Learning

Learning modules and progress tracking

Administration

Gesture, translation and learning-module management

Reporting

Report generation from stored data

No numerical accuracy/F1/response-time results are supplied in the source material, so none are fabricated here.

Performance Evaluation

Accuracy

Not measured in supplied material

Precision

Not measured

Recall

Not measured

F1-Score

Not measured

Response Time

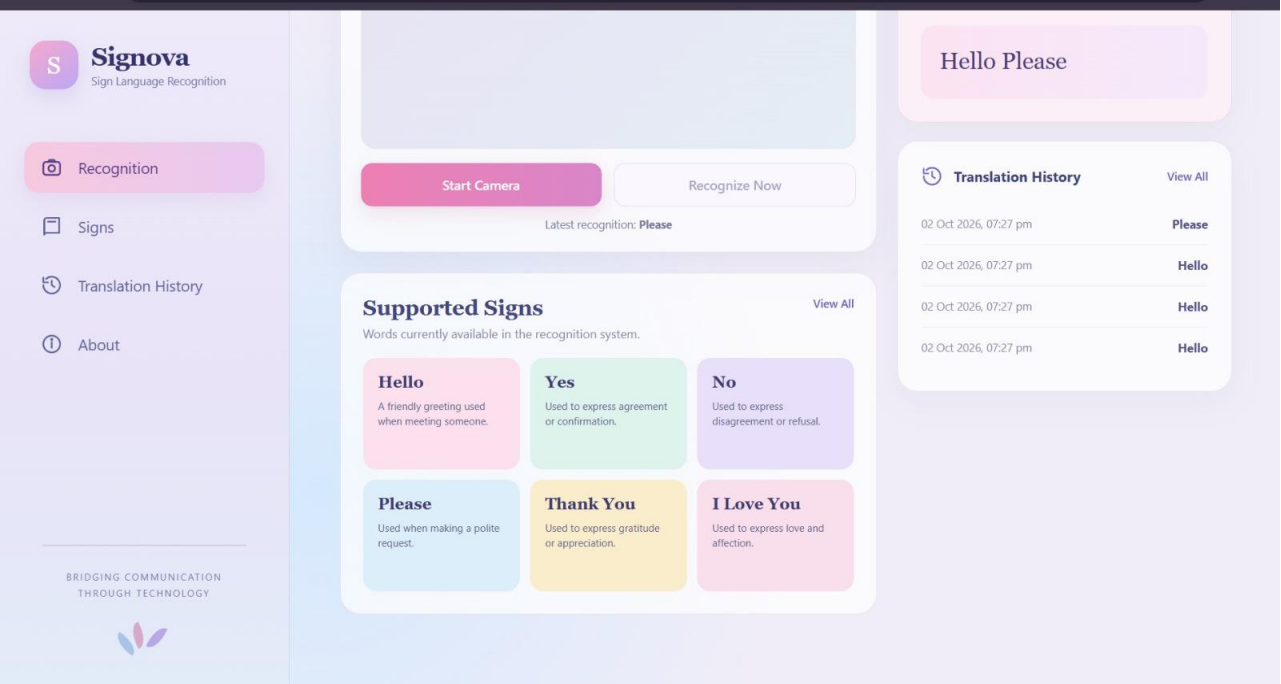
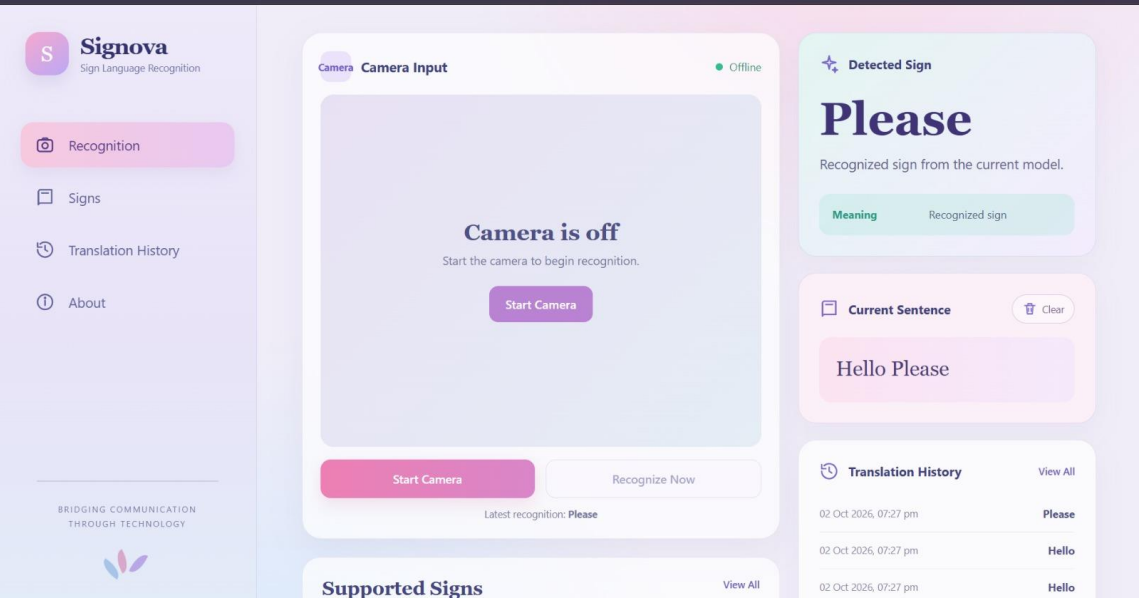
Not measured

Throughput

Not measured

For a future ML-based recognizer, these metrics can be measured on a labeled test set. For the current database application, functional correctness and response time are the more directly applicable evaluations.

Screenshots / Working Demo



Comparison with Existing System

Feature	Existing methods described	Proposed system
Centralized storage	Distributed across resources	MySQL-based centralized storage
Gesture search	Available through resources/apps	Integrated search
Translation history	Not identified as a common feature	Supported
Learning progress	Not identified as a common feature	Supported
Admin data management	Limited / not centralized	Add, update, delete records
DBMS structure	Not applicable across all resources	Keys, normalization, joins, triggers, procedures

Challenges & Solutions

CHALLENGE 1

Managing related user, gesture, translation and progress information.

SOLUTION

Use relational entities with primary/foreign keys and normalization.

CHALLENGE 2

Keeping translation history and learning progress organized.

SOLUTION

Dedicated Translation History and User Progress entities.

CHALLENGE 3

Supporting maintainable gesture and learning content.

SOLUTION

Admin operations for add/update/delete and extensible MySQL storage.

Future Work

Integrate computer vision and machine learning for automatic sign recognition.

Support camera-based gesture input instead of relying only on stored/reference sign information.

Expand the gesture and learning-content collection.

Evaluate the enhanced system using accuracy, precision, recall and F1 -score.

Improve usability, efficiency, and real-world deployment readiness.

Conclusion

The Sign Language Interpreter and Learning Management System provides a practical database-based approach for organizing sign-language information and supporting users in learning signs. Python provides the application interface and logic, while MySQL provides structured storage and retrieval. The design demonstrates DBMS concepts including database relationships, normalization, SQL queries, joins, triggers, stored procedures, and reporting. The project provides a foundation for future automatic sign recognition using computer vision and machine learning.

CONTRIBUTION

Centralized sign information + translation history + learning modules + progress management

References

References available from the supplied material:

Project Review – II PPT Template (provided template).

Sign Language Interpreter and Learning Management System project document (provided source PDF).

Sign-language reference visual included in the supplied project material.

Python documentation — to be added with official URL in the final submission.

MySQL documentation — to be added with official URL in the final submission.

Note: The supplied project PDF does not list individual research papers or a named dataset, so those entries should be added if required by your faculty.